# Electronic Supplementary Material for "A Component-Level Audit of No-Reference Quality Scoring in a Multi-Video Summarization Pipeline"

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This supplement holds the detailed breakdowns behind results reported in the article, relocated here to keep the article at a readable length. Nothing in it is new: every figure quoted in the article is reproduced in full below, and the article cites these tables as Table S1, Table S2, and so on.

**Table S1** In-domain collection: topics of Brazilian television news, with the number of source videos, segment slots, redundancy clusters and segment duration range. *Slots* counts cluster memberships; because two segments belong to two clusters each, the 33 slots correspond to 31 distinct segments.

| Topic | Videos | Slots | Clusters | Duration (s) | Status |
|---|---|---|---|---|---|
| bebe_real | 3 | 5 | 2 | 8–41 | Included |
| brumadinho | 3 | 6 | 3 | 13–188 | Included |
| fone_canc._ruido | 3 | 6 | 3 | 16–30 | Included |
| isencao_imposto | 3 | 6 | 3 | 25–30 | Included |
| mounjaro | 3 | 4 | 1 | 8–25 | Included |
| ucrania | 3 | 6 | 3 | 25–29 | Included |
| esquema_pcc | 3 | – | – | – | Excluded[†] |
| **Total** | **18** | **33** | **15** | 8–188 | — |

[†]98 SIFT descriptors $< K = 300$ $K$-means centroids. Slots exceed distinct segments in brumadinho and isencao_imposto, where one segment belongs to two clusters.

**Table S1 (continued). Operational 42-cluster set used in the primary application analysis.**

| Topic | Clusters | Slots | Distinct segments |
|---|---|---|---|
| acordo_mariana | 1 | 2 | 2 |
| aumento_cafe | 1 | 4 | 4 |
| bebe_real | 2 | 5 | 5 |
| bolsonaro_trump | 3 | 6 | 6 |
| brasil_omc | 1 | 2 | 2 |
| brumadinho | 3 | 6 | 5 |
| catedral_notre_dame | 2 | 5 | 4 |
| cessar_fogo_ucrania | 3 | 6 | 4 |
| esquema_pcc | 3 | 6 | 6 |
| fone_cancelamento_ruido | 3 | 6 | 6 |
| fraude_inss | 3 | 6 | 4 |
| google_huawei | 3 | 6 | 4 |
| isencao_imposto | 3 | 6 | 5 |
| julgamento_bolsonaro | 3 | 6 | 5 |
| mounjaro | 1 | 4 | 4 |
| recorde_calor_2024 | 2 | 5 | 5 |
| rio_celular | 1 | 2 | 2 |
| ucrania | 3 | 6 | 6 |
| ultraprocessados | 1 | 4 | 4 |
| **Total** | **42** | **93** | **83** |

**Table S2** Illustration of the sensitivity axis on a single broadcast frame and four degradations, after applying each method's declared polarity. The pattern this small case shows is the one Table S3 then measures over 25 distortion types and 10,125 images; nothing is concluded from this table alone. Cells are shaded by outcome: expected order, partial, violated.

| Method | Pol. | Original | Blur | Noise | Dark | JPEG | Outcome |
|---|---|---|---|---|---|---|---|
| QualiCLIP | ↑ | 0.341 | 0.155 | 0.167 | 0.196 | 0.163 | as expected |
| LIQE | ↑ | 1.294 | 1.056 | 1.000 | 1.023 | 1.023 | as expected |
| BRISQUE | ↓ | 42.8 | 90.0 | 66.4 | 59.1 | 78.2 | as expected |
| NIQE | ↓ | 4.53 | 11.99 | 15.79 | 6.18 | 9.47 | as expected |
| Keypoint density | ↑ | 2598 | 104 | **4543** | 4 | 2799 | partial |
| DINOv2 ViT-S/14 | ↑ | 14.69 | 14.86 | 14.86 | 14.90 | 15.14 | violated |
| ResNet-50 | ↑ | 15.62 | **18.11** | 15.00 | 15.97 | 15.28 | violated |

Values in bold are the ones that violate the expected order.

**Table S2 (continued). Exact parameters of the constructed broadcast degradations.**

| Type | Levels 1–5 |
|---|---|
| Gaussian blur | Kernel width $\{3, 5, 9, 15, 23\}$ pixels |
| White noise | Gaussian $\sigma = \{5, 12, 22, 35, 50\}$; deterministic seeds 1001–1005 |
| Darkening | RGB multiplier $\{.80, .62, .46, .32, .20\}$ with clipping |
| JPEG | OpenCV quality $\{40, 25, 15, 9, 5\}$ |
| H.264 | libx264 CRF $\{30, 36, 41, 45, 50\}$, preset `veryfast` |
| Rescale | Downscale factors $\{.75, .55, .40, .28, .18\}$, then restore original size |
| Motion blur | Horizontal kernel width $\{5, 9, 15, 23, 33\}$ pixels |
| Desaturation | HSV saturation multiplier $\{.75, .55, .35, .18, .05\}$ |

Each segment contributes at most 30 uniformly spaced frames. The seeded 20-segment simple random sample represents 13 topics: one segment each from `aumento_cafe`, `bolsonaro_trump`, `cessar_fogo_ucrania`, `esquema_pcc`, `isencao_imposto`, `mounjaro` and `rio_celular`; two each from `bebe_real`, `google_huawei`, `julgamento_bolsonaro`, `recorde_calor_2024` and `ucrania`; and three from `fone_cancelamento_ruido`.

**Table S3** Sensitivity to distortion level on KADID-10k. Direction-aligned SRCC is computed within each of the 81 reference images across five levels and then averaged by distortion type and across the 25 types. Intervals block-bootstrap the 81 references ($10^4$ resamples, seed 42). The last column counts types whose reference-averaged SRCC exceeds 0.50.

| Method | Mean $\rho^*$ | 95% block CI | Types $> 0.5$ |
|---|---|---|---|
| Q-Align | **0.880** | [0.874, 0.886] | **23/25** |
| QualiCLIP | **0.849** | [0.840, 0.857] | **22/25** |
| CLIP-IQA+ | **0.808** | [0.795, 0.821] | **22/25** |
| LIQE | **0.807** | [0.797, 0.817] | **22/25** |
| MUSIQ | **0.733** | [0.720, 0.747] | **20/25** |
| TOPIQ | **0.721** | [0.702, 0.737] | **20/25** |
| HyperIQA | 0.673 | [0.653, 0.692] | **19/25** |
| DBCNN | 0.668 | [0.642, 0.692] | **19/25** |
| ARNIQA† | 0.668 | [0.652, 0.683] | **20/25** |
| BRISQUE | 0.657 | [0.628, 0.684] | **19/25** |
| CLIP-IQA | 0.645 | [0.612, 0.676] | **20/25** |
| TReS† | 0.620 | [0.598, 0.640] | **18/25** |
| NIQE | 0.607 | [0.587, 0.625] | **19/25** |
| SI | 0.251 | [0.219, 0.282] | 13/25 |
| Keypoint density | 0.079 | [0.033, 0.128] | 7/25 |
| SIFT+BoVW | −0.077 | [−0.124, −0.030] | 5/25 |
| DINOv2 | −0.064 | [−0.136, 0.009] | 1/25 |
| ResNet-50 | −0.297 | [−0.364, −0.225] | 0/25 |

†ARNIQA and TReS are reported on this axis as coverage extensions but are not treated as complete-case methods in the screening rule because the required CLIVE evaluations were not available.

**Table S4** P.1401 point report after the logistic mapping. PLCC measures linearity, RMSE accuracy and the outlier ratio (OR; lower is better) consistency; RMSE* is the companion error after discounting the 95% confidence interval of the subjective score. OR saturates on KonIQ-10k because its per-image subjective intervals are narrow, but remains discriminative on KADID-10k. Analytic 95% intervals for PLCC, RMSE, RMSE* and OR are included in the released machine-readable results. A dagger marks evaluation on training data.

| Method | KonIQ-10k | | | | KADID-10k | | | |
|---|---|---|---|---|---|---|---|---|
| | PLCC | RMSE | RMSE* | OR | PLCC | RMSE | RMSE* | OR |
| Q-Align | .940 | 5.251 | 5.166 | .983 | .952 | .330 | .155 | .292 |
| LIQE | .915 | 6.234 | 6.149 | .983 | .720 | .751 | .526 | .703 |
| QualiCLIP | .771 | 9.837 | 9.752 | .990 | .673 | .800 | .572 | .725 |
| TOPIQ† | .978 | 3.198 | 3.124 | .959 | .546 | .907 | .671 | .769 |
| HyperIQA† | .956 | 4.507 | 4.428 | .977 | .503 | .936 | .700 | .776 |
| DBCNN† | .935 | 5.483 | 5.399 | .982 | .499 | .939 | .700 | .799 |
| MUSIQ† | .904 | 6.612 | 6.527 | .985 | .596 | .869 | .641 | .730 |
| CLIP-IQA+† | .863 | 7.790 | 7.705 | .987 | .665 | .809 | .581 | .727 |
| CLIP-IQA | .724 | 10.647 | 10.561 | .992 | .521 | .924 | .691 | .787 |
| SI | .561 | 12.782 | 12.696 | .993 | .328 | 1.023 | .783 | .821 |
| SIFT+BoVW (canon.) | .484 | 13.508 | 13.423 | .995 | .224 | 1.055 | .814 | .835 |
| Keypoint density | .484 | 13.502 | 13.417 | .994 | .227 | 1.055 | .814 | .836 |
| NIQE | .393 | 14.194 | 14.108 | .993 | .439 | .973 | .734 | .807 |
| BRISQUE | .253 | 14.931 | 14.844 | .995 | .395 | .995 | .758 | .807 |
| ResNet-50 | .162 | 15.231 | 15.143 | .994 | .194 | 1.062 | .821 | .834 |
| DINOv2 | .065 | 15.402 | 15.315 | .996 | .041 | 1.082 | .841 | .840 |

**Table S5** Mean direction-aligned SRCC by paradigm. The two blocks are separated because the change between databases means different things in each. In the upper block no member was fitted on either database, so Δ compares two generalization results. In the lower block at least one member was fitted on KonIQ-10k, marked †; there the KonIQ-10k column mixes fit with generalization and no Δ is meaningful.

| Paradigm | *N* | KADID | KonIQ | Δ | Best member |
|---|---|---|---|---|---|
| *Neither database seen by any member* | | | | | |
| ITU-T complexity | 1 | 0.219 | 0.513 | +0.294 | SI |
| SIFT+BoVW* | 2 | 0.124 | 0.446 | +0.322 | canonical IDF |
| Handcrafted | 1 | 0.123 | 0.447 | +0.324 | Keypoint density |
| NR-IQA, NSS | 2 | 0.355 | 0.302 | −0.053 | NIQE |
| CNN+BoVW | 19 | 0.318 | 0.058 | −0.260 | VGG-19+BoVW |
| Deep features | 2 | −0.086 | −0.115 | −0.029 | DINOv2 ViT-S/14 |
| *At least one member fitted on KonIQ-10k* | | | | | |
| NR-IQA, LMM | 1 | **0.951**† | **0.920**† | — | Q-Align |
| NR-IQA, CNN reg. | 2 | 0.473 | **0.933**† | — | HyperIQA |
| NR-IQA, transformer | 2 | 0.534 | **0.923**† | — | TOPIQ |
| NR-IQA, CLIP | 4 | 0.636 | **0.774**† | — | LIQE |

*Mean over the magnitudes of the two TF-IDF variants, which differ in orientation but not in strength. Restricted to the two CLIP members fitted on neither database, that paradigm gives 0.583 on KADID-10k and 0.679 on KonIQ-10k, a change of +0.096.

**Table S6** The BoVW variants in the domain, on the agreement axis and on what they deliver. *Differs* counts the clusters in which the variant selects a different segment from the deployed one; *Agreement* is top-1 agreement with the reference model over the 15 clusters; $r_{\mathrm{kp}}$ is the rank correlation with keypoint density over the 33 segment slots; and the last three columns are the summary the variant delivers, as in Table S11. The comparison between summing and averaging defined in the main article is therefore evaluated directly. The final column is the uncalibrated raw sum of the DOVER technical and aesthetic branches, used only as a scale-sensitivity diagnostic.

| Variant | Aggr. | Differs | Agreem. | $r_{\mathrm{kp}}$ | Dur. (s) | Q-Align | raw T+A |
|---|---|---|---|---|---|---|---|
| SIFT+BoVW (as implemented) | sum | — | 9/15 | +0.591 | 634.0 | 61.1% | 52.1% |
| SIFT+BoVW (canonical IDF) | sum | 1/15 | 10/15 | +0.116 | 801.5 | 62.2% | 47.6% |
| BoVW / descriptor count | mean | 2/15 | 10/15 | −0.199 | 810.9 | 59.0% | 48.2% |
| **BoVW / frame count** | mean | 1/15 | 10/15 | −0.145 | **459.3** | **77.3%** | **71.5%** |
| Keypoint density | mean | 5/15 | 7/15 | +1.000 | — | — | — |
| **LIQE** (for reference) | mean | 5/15 | 11/15 | −0.381 | 455.6 | 85.8% | 91.8% |

**Table S7** Descriptive in-domain agreement with the Q-Align reference model over the 15 human-annotated redundancy clusters, raw and chance-corrected against $\bar{p}_e = 0.472$. These clusters share topics and two pairs share a segment; the table therefore makes no cluster-independence inference. Topic-block intervals for the larger system-defined set are reported in the main article. Roles follow Table 1: D = deployed, C = candidate and X = diagnostic.

| Method | Role | Agreement | $\kappa_{\mathrm{uniform}}$ |
|---|---|---|---|
| MUSIQ | C | 13/15 (86.7%) | +0.747 |
| DBCNN | C | 12/15 (80.0%) | +0.621 |
| TOPIQ | C | 12/15 (80.0%) | +0.621 |
| LIQE | C | 11/15 (73.3%) | +0.495 |
| HyperIQA | C | 11/15 (73.3%) | +0.495 |
| NIQE | C | 10/15 (66.7%) | +0.368 |
| QualiCLIP | C | 10/15 (66.7%) | +0.368 |
| BRISQUE | C | 10/15 (66.7%) | +0.368 |
| SIFT+BoVW | D | 9/15 (60.0%) | +0.242 |
| CLIP-IQA | C | 8/15 (53.3%) | +0.116 |
| CLIP-IQA+ | C | 8/15 (53.3%) | +0.116 |
| DINOv2 | X | 5/15 (33.3%) | −0.263 |

**Table S8** The four redundancy clusters in which the method under study diverges from the majority of the quality references. Each row is a candidate segment; each column a scoring method. The top-1 choice of each method is shaded: pink for SIFT+BoVW, green for the quality references. NIQE and BRISQUE are lower-is-better. Scores are reproduced from the aggregated per-cluster results.

| Segment | SIFT+BoVW | LIQE | NIQE↓ | Q-Align | QualiCLIP | BRISQUE↓ |
|---|---|---|---|---|---|---|
| *bebe_real C1* | | | | | | |
| Band 0:37 (10 s) | **5448.0** | 1.49 | 6.55 | 4.02 | 0.28 | 50.97 |
| Record 0:46 (11 s) | 2077.6 | **1.59** | **5.57** | **4.17** | **0.30** | **41.44** |
| *brumadinho C2* | | | | | | |
| Band 0:00 (13 s) | 2888.3 | **2.60** | **5.04** | **4.84** | **0.36** | **44.95** |
| Record 4:24 (188 s) | **5332.7** | 1.13 | 6.23 | 3.74 | 0.27 | 53.57 |
| *isencao_imposto C0* | | | | | | |
| Band 2:46 (30 s) | **3732.8** | 4.23 | **4.67** | 3.70 | **0.62** | 34.65 |
| CNN 1:12 (25 s) | 3010.0 | **4.47** | 4.74 | **4.26** | 0.59 | **29.84** |
| *mounjaro C0* | | | | | | |
| Band 0:00 (8 s) | 559.8 | 4.12 | 7.29 | **4.38** | 0.56 | 63.19 |
| Dom. Esp. 2:06 (16 s) | **1945.1** | 3.95 | 5.87 | 3.90 | 0.57 | 36.55 |
| Dom. Esp. 3:07 (25 s) | 1784.4 | 3.61 | 5.28 | 3.68 | 0.56 | 42.29 |
| Record 0:00 (14 s) | 815.2 | **4.93** | **4.76** | 4.26 | **0.63** | **35.45** |

**Table S9** Top-1 agreement with the reference model over the 15 clusters under four temporal poolings. The worst-frame, tenth-percentile and mean − sd rules all penalise within-segment dispersion; the mean ignores it.

| Method | Mean | Worst frame | 10th pct. | mean − sd |
|---|---|---|---|---|
| **LIQE** | 11/15 | 11/15 | 11/15 | 11/15 |
| NIQE | 10/15 | 10/15 | 10/15 | 10/15 |
| QualiCLIP | 9/15 | 9/15 | 11/15 | 10/15 |
| BRISQUE | 10/15 | 9/15 | 9/15 | 10/15 |

**Table S10** Eligible set as a function of the generalization threshold, under both readings of "a database it was not fitted on". *Universal* requires the threshold to hold on every unseen database (the reading adopted in the main article); *existential* requires it on at least one. The sensitivity requirement is held at a within-reference mean above 0.50 and at least 15/25 types; the topic-block lower confidence limit must also exceed the in-domain floor.

| Minimum $\rho^*$ | Universal reading (adopted) | Existential reading |
|---|---|---|
| 0.50 | LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, CLIP-IQA | LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, CLIP-IQA, DBCNN, HyperIQA |
| 0.55 | LIQE, QualiCLIP, CLIP-IQA+, MUSIQ | *same eight methods* |
| 0.60 | **LIQE, QualiCLIP, CLIP-IQA+** | *same eight methods* |
| 0.65* | **LIQE, QualiCLIP, CLIP-IQA+** | *same eight methods* |
| 0.70 | LIQE | *same eight methods* |
| 0.75 | none | LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, DBCNN, HyperIQA |

*The value used. At the adopted generalization threshold and universal reading, the same three methods survive for sensitivity type-count cuts from 12/25 through 22/25; none survives at 23/25. Requiring only a positive floor point estimate instead of a topic-block lower limit does not change the three-method set.

**Table S11** The same comparison on the 15 human-annotated clusters. The two references agree closely here; their disagreement emerges on the 42-cluster analysis in the main article, which is the primary estimate. The DOVER column uses the uncalibrated raw technical-plus-aesthetic sum, not DOVER-T.

| Scoring method | Changed | by raw T+A | by Q-Align | Dur. (s) | Words |
|---|---|---|---|---|---|
| SIFT+BoVW (deployed) | — | 52.1% | 61.1% | 634.0 | 1,613 |
| **LIQE** | 5/15 | 91.8% | 85.8% | 455.6 | 1,124 |
| BRISQUE | 6/15 | 84.5% | 84.7% | 431.0 | 1,073 |
| QualiCLIP | 4/15 | 80.9% | 80.4% | 439.6 | 1,124 |
| NIQE | 4/15 | 80.9% | 80.4% | 439.6 | 1,124 |
| DINOv2 | 12/15 | 53.4% | 36.7% | 328.6 | 904 |

**Table S11 (continued). Construct and scale sensitivity on the 42 operational clusters.**

| DOVER output | Estimand | SIFT | LIQE | Δ; topic CI | Hierarchical CI |
|---|---|---|---|---|---|
| Technical branch | Global range (%) | 50.5 | 76.0 | +25.5; [−12.1, 54.9] | [−17.2, 59.6] |
| Technical branch | Raw mean | −.0770 | −.0701 | +.0069; [−.0028, .0178] | [−.0041, .0197] |
| Technical branch | Mean cluster position (%) | 57.1 | 64.4 | +7.2; [−20.7, 35.1] | [−22.4, 37.5] |
| Technical branch | Selected maximum (%) | 54.8 | 59.5 | +4.8; [−22.2, 31.7] | [−24.4, 34.2] |
| Raw T+A sum | Global range (%) | 47.4 | 77.7 | +30.3; [1.1, 54.1] | [−4.1, 59.0] |
| Aesthetic branch | Global range (%) | 45.0 | 73.7 | +28.7; [5.3, 49.7] | [−.2, 54.6] |

The branch sum is the raw technical output plus the raw aesthetic output. It is not the official ITU-standardized DOVER fused score. The technical branch is primary; the other rows diagnose construct sensitivity.

| DOVER-T range filter | Clusters | Threshold | Global-range Δ |
|---|---|---|---|
| All clusters | 42 | — | +25.5 |
| Exclude bottom 10% | 37 | .00397 | +25.9 |
| Exclude bottom 25% | 31 | .00891 | +28.1 |

DOVER-T cluster ranges: minimum .00028, first quartile .00891, median .01686, third quartile .04118, maximum .10653; zero-range clusters: 0.

**Matched-input control: ten identical frames and mean pooling.**

| Scorer | Differs from SIFT | DOVER-T range | Raw mean | Selected max |
|---|---|---|---|---|
| SIFT+BoVW | — | 52.0% | −.0766 | 57.1% |
| LIQE | 16/42 | 79.1% | −.0692 | 64.3% |

LIQE–SIFT global-range Δ = +27.1 points (topic CI [5.6, 47.8]; hierarchical CI [2.4, 52.5]). Raw-branch Δ = +.00736 (topic [.00135, .01406]; hierarchical [.00057, .01683]). Thirteen of the 16 disagreement clusters belong to the 22-cluster complete-operation change set. Relative to the published deployed path, matched-input SIFT changes 12 choices and matched-input LIQE changes 22; the control isolates scorer identity, not the historical operation itself.

**Table S12** The eight predictions registered at commit `9377c07` before the constructed experiment, against the measurements. None is used to filter a method.

| | Subject | Prediction | Measured | Outcome |
|---|---|---|---|---|
| P1 | SIFT+BoVW | High accuracy on blur and darkening | 90%, 100% | holds |
| P2 | SIFT+BoVW | Inverted on white noise | 8% | holds |
| P3 | SIFT+BoVW | Near chance on JPEG and H.264 | 6% and 90% | refuted |
| P4 | CNN+BoVW (19) | Chance on every type | 44.9%, CI [41.0, 48.9]; range 30–64 | refuted |
| P5 | NIQE, BRISQUE | Accuracy falls under heavy overlay | +2.0, +2.2 points | refuted |
| P6 | Learned methods | No interaction with overlay | −18.1 to −8.1 for three | refuted |
| P7 | SIFT+BoVW | Design B mean below 0.15 | −0.425 | refuted |
| P8 | Q-Align | Design A accuracy above 0.90 | 98.2%, CI [97.2, 99.1] | holds |

The complete registered commit is:

`9377c07ca080ac61c7cff80e0ff6c7507b6ebaba`.

The submission archive contains `registered_prediction_commit.bundle`; the clone, bundle verification and object-inspection commands in its README make the commit and timestamp independently inspectable even before the public release.

**Table S12 (continued). Fidelity and descriptor-count diagnostics for the constructed broadcast pairs.**

| Method | All 8 types (800) | | | Without dark/desaturation (600) | | |
|---|---|---|---|---|---|---|
| | Correct | Ties | Accuracy | Correct | Ties | Accuracy |
| SIFT+BoVW | 521 | 0 | 65.1% | 382 | 0 | 63.7% |
| LIQE | 796 | 0 | 99.5% | 599 | 0 | 99.8% |
| Q-Align | 786 | 0 | 98.2% | 600 | 0 | 100.0% |
| DOVER-T | 781 | 0 | 97.6% | 599 | 0 | 99.8% |

| Type | Mean $\Delta D$ | $\Delta D > 0$ | Mean $\Delta D$ by level 1–5 |
|---|---|---|---|
| Blur | −797 | 10% | −130, −417, −913, −1188, −1339 |
| Dark | −796 | 0% | −229, −487, −791, −1116, −1358 |
| H.264 | −398 | 10% | −128, −220, −361, −541, −737 |
| JPEG | +289 | 95% | +161, +252, +339, +376, +317 |
| Motion blur | −1067 | 0% | −548, −909, −1174, −1308, −1396 |
| Noise | +469 | 92% | +126, +348, +517, +607, +749 |
| Rescale | −646 | 12% | −154, −377, −585, −911, −1204 |
| Desaturation | +18 | 61% | 0, +14, +21, +34, +24 |

$\Delta D = D_{\text{degraded}} - D_{\text{unmodified}}$. Strict accuracy uses correct/total; a tie is neither correct nor incorrect and remains in the denominator.

| DOVER-T | dark | motion blur | H.264 | rescale | satur. | noise | JPEG | All |
|---|---|---|---|---|---|---|---|---|
| Accuracy (%) | 93 | 100 | 100 | 100 | 100 | 89 | 99 | 100 | 97.6 |
| Mean severity $\rho^*$ | .985 | .900 | 1.000 | 1.000 | 1.000 | .715 | .995 | 1.000 | — |

The raw technical output is the exact branch later used for the 42-cluster output analysis. Its overall segment-block 95% CI is [96.1, 99.0]; after excluding darkening and desaturation, accuracy is 99.8% with CI [99.5, 100]. There are no ties. Each condition is encoded through the same 5-fps MP4 wrapper and uses paired spatial fragment seed 42.

**Table S13** Method inventory and released-weight supervision. *Sup.*: OU = opinion-unaware, K = KonIQ-10k, L = LIVE, M = multi-database mixture. Roles follow Table 1: D = deployed, C = candidate, R = reference and X = diagnostic or ablation. The daggered partial-coverage methods are excluded from complete-case screening.

| Method | Family | Construct | Sup. | Pol. | GPU | Role |
|---|---|---|---|---|---|---|
| SIFT+BoVW | Handcrafted | TF-IDF word count | – | ↑ | No | D |
| Keypoint density | Handcrafted | keypoints/frame | – | ↑ | No | X |
| DINOv2 | Self-supervised | patch-token norm | – | ↑ | Yes | X |
| ResNet-50 | Supervised | pooled-feature norm | – | ↑ | Yes | X |
| BRISQUE | NSS | natural-statistics deviation | L | ↓ | No | C |
| NIQE | NSS | pristine-model distance | OU | ↓ | No | C |
| QualiCLIP | CLIP | quality score | OU | ↑ | Yes | C |
| LIQE | CLIP | quality, distortion, scene | K | ↑ | Yes | C |
| Q-Align | LMM | textual quality levels | M | ↑ | Yes | R |
| SI | ITU-T complexity | Sobel energy | – | ↑ | No | X |
| 19 CNN+BoVW | ablation | TF-IDF word count | – | ↑ | Yes | X |
| DBCNN | CNN regression | bilinear pooling | K | ↑ | Yes | C |
| HyperIQA | CNN regression | adaptive weights | K | ↑ | Yes | C |
| MUSIQ | Transformer | multi-scale score | K | ↑ | Yes | C |
| TOPIQ | Transformer | semantic attention | K | ↑ | Yes | C |
| CLIP-IQA | CLIP | antonym prompts | OU | ↑ | Yes | C |
| CLIP-IQA+ | CLIP | tuned prompts | K | ↑ | Yes | C |
| ARNIQA† | Self-supervised | distortion manifold | K | ↑ | Yes | partial |
| TReS† | CNN–transformer | ranking/consistency | K | ↑ | Yes | partial |

PyIQA registry keys used by the archived scorers are `brisque`, `niqe`, `dbcnn`, `hyperiqa`, `musiq`, `topiq_nr`, `clipiqa`, `clipiqa+`, `liqe`, `qualiclip`, `arniqa` and `tres`. The final inventory and analysis archive were frozen on 23 August 2026. MANIQA was attempted with PyIQA 0.1.15.post2 and the `maniqa-kadid` checkpoint. A single-image diagnostic completed, but the released implementation evaluates 20 crops per image and the full 21,360-image three-database run did not finish within the experiment window on the single 8-GB RTX 4060; it is therefore an uncompleted method, not a failed candidate.

**Table S13** (continued). Complete extractor inventory for the CNN+BoVW ablation. Every row uses the final spatial tensor before global pooling or classification.

| Extractor | Released torchvision weights | Spatial endpoint |
|---|---|---|
| ResNet-18 | IMAGENET1K_V1 | trunk before average pooling |
| ResNet-50 | IMAGENET1K_V2 | trunk before average pooling |
| ResNet-101 | IMAGENET1K_V2 | trunk before average pooling |
| VGG-16 | IMAGENET1K_V1 | full `features` module |
| VGG-19 | IMAGENET1K_V1 | full `features` module |
| EfficientNet-B0 | IMAGENET1K_V1 | full `features` module |
| MobileNet-V3-Large | IMAGENET1K_V2 | full `features` module |
| ConvNeXt-Tiny | IMAGENET1K_V1 | full `features` module |
| ConvNeXt-Small | IMAGENET1K_V1 | full `features` module |
| DenseNet-121 | IMAGENET1K_V1 | full `features` module |
| DenseNet-201 | IMAGENET1K_V1 | full `features` module |
| ResNeXt-50-32x4d | IMAGENET1K_V2 | trunk before average pooling |
| ResNeXt-101-64x4d | IMAGENET1K_V1 | trunk before average pooling |
| Wide-ResNet-50-2 | IMAGENET1K_V2 | trunk before average pooling |
| Wide-ResNet-101-2 | IMAGENET1K_V2 | trunk before average pooling |
| Inception-V3 | IMAGENET1K_V1 | convolutional trunk through Mixed_7c |
| EfficientNet-B4 | IMAGENET1K_V1 | full `features` module |
| SqueezeNet-1.1 | IMAGENET1K_V1 | full `features` module |
| MobileNet-V2 | IMAGENET1K_V2 | full `features` module |

Common preprocessing: bicubic resize to 256 pixels, center crop to $224 \times 224$, ImageNet mean and standard deviation, and FP16 inference. Each spatial location is flattened into one descriptor and $\ell_2$-normalized in FP32. The same internal key-frame filter, vocabulary construction, TF–IDF weighting and segment aggregation are then applied. The archived implementation is `cnn_bovw_scorer.py`.

**Table S13** (continued). Exact configurations of the two model-based references used in the broadcast analyses.

| Model | Config. | Input / pooling | Output used |
|---|---|---|---|
| Q-Align | `q-future/one-align`; task `quality`, input `image` | Uniform 1-fps frames; arithmetic mean of frame scores | Four-bit NF4 with double quantisation and FP16 compute; higher score is better. The same configuration scores the 800 constructed pairs and supplies the in-domain floor. |
| DOVER | Official `DOVER.pth` and `dover.yml`, `val-l1080p` view decomposition | Technical: $7 \times 7$ spatial fragments of size 32, 32 frames, interval 2, three clips; aesthetic: $224 \times 224$, 32 frames, interval 2, one clip | The technical branch is the primary output reference and is also evaluated on the 800 pairs with a common 5-fps MP4 wrapper and paired seed 42. The raw branch sum is only a sensitivity analysis; it is not the official standardised fusion or DOVER++. |

**Table S14** Computational cost measured on one RTX 4060 over the same inputs. Frame-wise entries report median latency after ten warm-up and sixty timed iterations; the final two rows are measured complete paths.

| Method | ms/frame | IQR | Per cluster |
|---|---|---|---|
| SI | 2.2 | 0.14 | 0.2 s |
| QualiCLIP | 14.3 | 1.68 | 1.2 s |
| TOPIQ | 18.7 | 0.36 | 1.5 s |
| BRISQUE | 19.6 | 0.98 | 1.6 s |
| CLIP-IQA+ | 21.4 | 0.65 | 1.8 s |
| DBCNN | 21.5 | 0.46 | 1.8 s |
| LIQE | 22.7 | 0.82 | 1.9 s |
| CLIP-IQA | 25.2 | 0.51 | 2.1 s |
| NIQE | 34.6 | 0.90 | 2.8 s |
| MUSIQ | 40.0 | 9.36 | 3.3 s |
| ResNet-50 | 46.2 | 8.44 | 3.8 s |
| DINOv2 | 46.4 | 36.63 | 3.8 s |
| Keypoint density | 46.8 | 1.48 | 3.8 s |
| HyperIQA | 50.1 | 0.77 | 4.1 s |
| SIFT+BoVW | $\sim$ 1,724 | — | 182 s |
| Q-Align | 322.6 | — | 26 s |

The uniform 1-fps candidate cache contains 2,892 frames from 83 distinct segments. Counting repeated memberships across the 93 cluster slots gives 3,525 segment-frame instances. Candidate latency was timed per decoded frame; the SIFT+BoVW row is the complete operation divided by its input-frame count and therefore includes key-frame selection, descriptor extraction, vocabulary quantisation and aggregation. The per-cluster column is the operational quantity; the per-frame SIFT value is an approximate normalization rather than an identical-path microbenchmark.

**Table S15** Descriptive three-database contrasts and released-weight exposure. K–C is KonIQ-10k minus CLIVE; C–KADID is CLIVE minus KADID-10k. Positive values are losses. K = KonIQ-10k opinion training, OU = opinion-unaware and L = LIVE opinion training. The contrasts do not isolate causal effects because the databases, architectures and objectives are unmatched.

| Method | Sup. | KonIQ | CLIVE | KADID | K–C | C–KADID |
|---|---|---|---|---|---|---|
| *Released checkpoint fitted on KonIQ-10k* | | | | | | |
| TOPIQ | K | 0.973† | 0.811 | 0.511 | +0.162 | +0.300 |
| HyperIQA | K | 0.947† | 0.750 | 0.462 | +0.197 | +0.288 |
| DBCNN | K | 0.919† | 0.756 | 0.484 | +0.163 | +0.272 |
| MUSIQ | K | 0.873† | 0.789 | 0.556 | +0.084 | +0.233 |
| CLIP-IQA+ | K | 0.824† | 0.805 | 0.654 | +0.019 | +0.151 |
| LIQE | K | 0.915† | 0.865 | 0.723 | +0.050 | +0.142 |
| *mean over the six* | | | | | +0.113 | +0.231 |
| *Never fitted on KonIQ-10k* | | | | | | |
| QualiCLIP | OU | 0.711 | 0.757 | 0.665 | −0.046 | +0.092 |
| CLIP-IQA | OU | 0.647 | 0.703 | 0.501 | −0.056 | +0.202 |
| NIQE | OU | 0.368 | 0.449 | 0.379 | −0.081 | +0.070 |
| BRISQUE | L | 0.236 | 0.313 | 0.330 | −0.077 | −0.017 |
| *mean over the four* | | | | | −0.065 | +0.087 |

†In-distribution. CLIVE contains 1,162 images and is unseen by every checkpoint in the table. The operational partition assigns 33% of the mean KonIQ-to-KADID gap of the fitted group to K–C and 67% to C–KADID; these percentages describe contrasts and are not causal estimates.

**Table S16** Spearman association with human scores before and after conditioning on detectable keypoint density. $r_{\mathrm{kp}}$ is the rank correlation between the method score and keypoint density. Differences are descriptive attenuations, not causal fractions of "content density."

| Method | KonIQ-10k | | | | KADID-10k | | | |
|---|---|---|---|---|---|---|---|---|
| | $r_{\mathrm{kp}}$ | $\lvert\rho\rvert$ | $\lvert\rho_{\cdot\mathrm{kp}}\rvert$ | Diff. | $r_{\mathrm{kp}}$ | $\lvert\rho\rvert$ | $\lvert\rho_{\cdot\mathrm{kp}}\rvert$ | Diff. |
| Q-Align | +.44 | .920 | .901 | −.019 | +.13 | .951 | .950 | −.001 |
| LIQE | +.43 | .915 | .894 | −.020 | +.18 | .723 | .718 | −.005 |
| TOPIQ | +.48 | .973 | .966 | −.007 | +.21 | .511 | .500 | −.011 |
| HyperIQA | +.48 | .947 | .933 | −.014 | +.17 | .462 | .451 | −.011 |
| DBCNN | +.49 | .919 | .898 | −.021 | +.22 | .484 | .472 | −.012 |
| MUSIQ | +.44 | .873 | .841 | −.032 | +.20 | .556 | .546 | −.010 |
| QualiCLIP | +.53 | .711 | .626 | −.086 | +.29 | .665 | .663 | −.002 |
| CLIP-IQA | +.27 | .647 | .611 | −.036 | +.22 | .501 | .489 | −.012 |
| SI | +.77 | .513 | .296 | −.218 | +.70 | .219 | .187 | −.032 |
| NIQE | −.63 | .368 | .125 | −.243 | −.33 | .379 | .362 | −.018 |
| BRISQUE | −.31 | .236 | .116 | −.120 | −.13 | .330 | .319 | −.011 |
| SIFT+BoVW (canonical) | +.998 | .447 | .016 | arith. | +.992 | .125 | .028 | arith. |
| SIFT+BoVW (deployed) | −.9999 | .445 | n.d. | arith. | −.9998 | .123 | n.d. | arith. |

The shaded rows are the near-tautological regime: conditioning a score on a variable with which it is almost perfectly correlated removes the variable from itself. For the deployed row the partial coefficient is not reported because the denominator approaches zero and the result is numerically unstable.

**Table S17** The sixteen operations audited, each linked to the objective its source states and the role it has there. *Tested reading* is the one this article evaluates: higher score means better technical quality, applied to a single frame without a model. No source is claimed to be wrong about its own objective; what is tested is the reuse. *Inverts* counts degradation types on which the operation ranks the degraded copy higher (Fig. 2 of the main article). Where two works adopt the same operation the use can differ: Laplacian variance rejects low-light frames alongside grey-scale intensity and a Canny detector in Cakmak et al. [1], and enters $0.7\,z(\text{sharp}) + 0.3\,z(\text{bright})$ in Korolkov [8].

| Operation | Objective in the source | Role there | Inverts |
|---|---|---|---|
| Averaged aesthetics (3) [2] | Keyframe interestingness | unweighted mean of three aesthetic terms; the source gives no weights | **1**/8 |
| Colourfulness (Hasler) [3] | Perceived colourfulness of a scene | stand-alone perceptual index | **1**/8 |
| Colourfulness (Ke) [7] | Professional vs. snapshot photograph | feature of a naive Bayes classifier | **1**/8 |
| Composite of five [10] | Keyframe ranking | unweighted mean of five features | **1**/8 |
| Contrast [7] | Professional vs. snapshot photograph | feature of a naive Bayes classifier | **2**/8 |
| Contrast variance [6] | Static-frame selection | one of four inputs to a cost multiplicative in cluster distance | **2**/8 |
| Edge distribution [7] | Professional vs. snapshot photograph | feature of a naive Bayes classifier | **2**/8 |
| Entropy [9] | Keyframe diversity and coverage | SE in segment coverage (Eq. 21) | **2**/8 |
| GLCM contrast [12] | Keyframe extraction | texture descriptor | **1**/8 |
| Gradient magnitude [9] | Keyframe diversity and coverage | GM in visual attention (Eq. 20) | **3**/8 |
| Gradient orientation [9] | Keyframe diversity and coverage | GO in segment diversity (Eq. 22) | **4**/8 |
| Hue count [7] | Professional vs. snapshot photograph | feature of a naive Bayes classifier | **5**/8 |
| Laplacian variance [1] | Blur rejection in frame filtering | sharpness threshold | **1**/8 |
| Saliency area [9] | Keyframe diversity and coverage | VS in visual attention (Eq. 20) | **6**/8 |
| Saturation [4] | Aesthetic scoring | one term among several | **3**/8 |
| Spatial information [5] | Spatial detail of a test sequence | characterization, not quality | — |

*Tested reading, identical for all sixteen*: the operation is computed on one frame and read as a technical-quality score in which a higher value marks the better of two versions of the same content. Fifteen are closed-form; saliency area requires a learned map in its source and is marked as such.

**Table S18** The sixteen visual-selection operations the video-summarization literature uses at the selection stage, reimplemented from the published equations and measured under one protocol. *Sensitivity* is the direction-aligned SRCC within each KADID-10k reference, averaged over the 25 distortion types, with the count of types above .50; *Generalization* is the lowest such correlation on any database the method was not fitted on; *Floor* is chance-corrected agreement on the operational clusters. **The two families use different reference models there**: operations against LIQE, the ten no-reference models against Q-Align, so that column is not a comparison between families. The damage columns share one protocol. The rule admits a method when sensitivity exceeds .5 on at least 15 of 25 types and generalization exceeds .65. **No operation clears either damage gate**: the best sensitivity is .298 and the best generalization among the fifteen evaluated is .139. Spatial information declares no direction, carries no verdict and is excluded from every count. The *Floor* column reports the in-domain agreement against the reference model with the lower end of its 95% interval in brackets, so a pass can be read for how much of it survives the interval: the weakest passing operation in this table has a lower bound of +.011.

| Operation | Source | Sens. | Types | Gen. | Floor $\kappa$ [95% lo] | Elig. |
|---|---|---|---|---|---|---|
| Edge distribution | [7] | .298 | 9/25 | +.083 | −.258 [−.484] | no |
| Gradient orientation | [9] | .291 | 12/25 | −.035 | — | no |
| Spatial information | [5] | .251 | 13/25 | +.218 | +.506 [+.280] | — |
| Contrast | [7] | .185 | 10/25 | +.139 | +.371 [+.106] | no |
| Contrast variance | [6] | .149 | 11/25 | +.096 | +.101 [−.157] | no |
| Hue count | [7] | .147 | 6/25 | −.162 | −.213 [−.474] | no |
| Entropy | [9] | .117 | 11/25 | +.102 | — | no |
| Gradient magnitude | [9] | .104 | 11/25 | +.129 | — | no |
| Saliency area | [9] | .075 | 3/25 | +.031 | — | no |
| Averaged aesthetics (3) | [2] | .032 | 8/25 | +.027 | — | no |
| Composite of five | [10] | .029 | 11/25 | −.020 | +.416 [+.160] | no |
| GLCM contrast | [12] | .019 | 9/25 | +.136 | — | no |
| Laplacian variance | [1] | −.047 | 9/25 | +.095 | +.551 [+.328] | no |
| Saturation | [4] | −.092 | 7/25 | −.254 | −.034 [−.325] | no |
| Colourfulness (Hasler) | [3] | −.182 | 4/25 | −.062 | +.281 [+.011] | no |
| Colourfulness (Ke) | [7] | −.182 | 4/25 | −.062 | +.281 [+.012] | no |

**Table S19** Which operation entered which test. A filled marker means the operation has a result on that axis. **Coverage is not uniform, and the two reasons differ.** Spatial information carries no paired verdict because its source declares no direction, which is a property of the source. The seven operations without a downstream result were added to the implementation after the downstream pass was run, which is a property of our sequence of work and not of the method; extending that pass to them is a stated next step. Sensitivity and generalization cover all 16. Counts per axis are in the last row.

| Operation | Paired | Sens. | Gen. | Floor | Down. | Absent because |
|---|---|---|---|---|---|---|
| Averaged aesthetics (3) | ● | ● | ● | ○ | ○ | after the downstream pass |
| Colourfulness (Hasler) | ● | ● | ● | ● | ● | |
| Colourfulness (Ke) | ● | ● | ● | ● | ● | |
| Composite of five | ● | ● | ● | ● | ● | |
| Contrast | ● | ● | ● | ● | ● | |
| Contrast variance | ● | ● | ● | ● | ● | |
| Edge distribution | ● | ● | ● | ● | ● | |
| Entropy | ● | ● | ● | ○ | ○ | after the downstream pass |
| GLCM contrast | ● | ● | ● | ○ | ○ | after the downstream pass |
| Gradient magnitude | ● | ● | ● | ○ | ○ | after the downstream pass |
| Gradient orientation | ● | ● | ● | ○ | ○ | after the downstream pass |
| Hue count | ● | ● | ● | ● | ● | |
| Laplacian variance | ● | ● | ● | ● | ● | |
| Saliency area | ● | ● | ● | ○ | ○ | after the downstream pass |
| Saturation | ● | ● | ● | ● | ● | |
| Spatial information | ○ | ● | ● | ● | ○ | source declares no direction |
| **of 16** | **15** | **16** | **16** | **10** | **9** | |

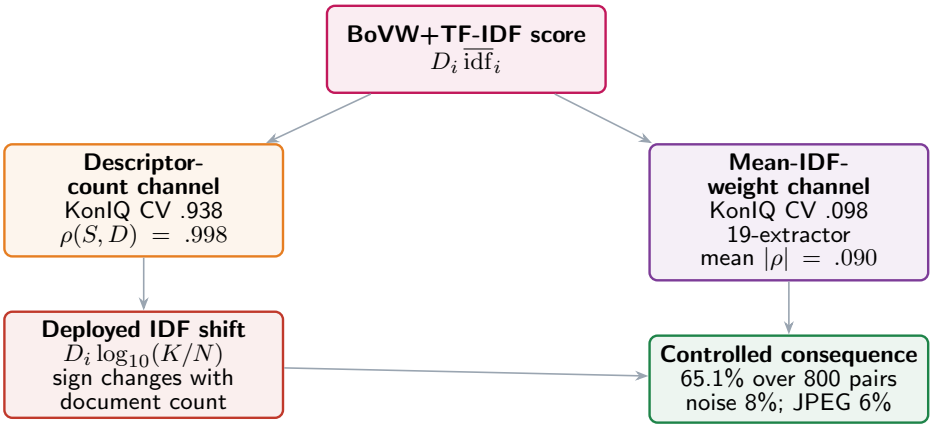



**Fig. S1** Measured mechanism. The factorization identifies two channels; the data determine their relative variation and connect count dominance to the controlled-degradation failures.

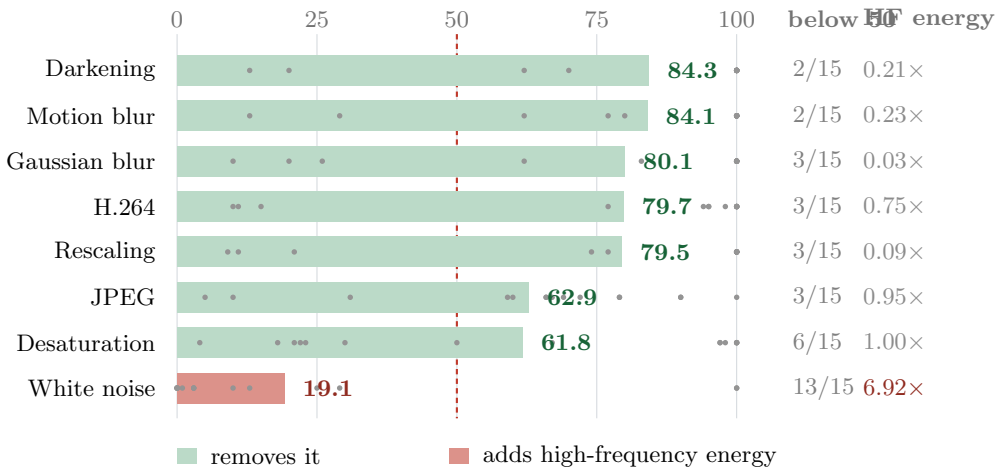



**Fig. S2** Mean paired accuracy of the fifteen summarization operations, by degradation type; grey dots are the individual operations and the dashed line is chance. The ordering follows what each degradation does to local image high-frequency energy, measured as the median ratio to the original and shown in the last column, rather than how severe the degradation is. Seven of the eight reduce that energy and are answered correctly. **White noise is the only one that raises it**, by a factor of 6.9, and it is answered at 19.1 per cent with a median of 3.0. The middle column counts operations below chance.

**Table S20** Model-reference output over the 42 operational redundancy clusters. Changed counts representatives differing from the deployed operation. DOVER-T is the raw DOVER technical branch and is not used in screening; Q-Align defines the in-domain screening floor. Both outputs use the global-achievable-range estimand of Eq. 3 of the main article; Max is the percentage of clusters in which the selected representative has the highest DOVER-T score.

| Quality operation | Changed | DOVER-T | Max | Q-Align |
|---|---|---|---|---|
| SIFT+BoVW | — | 50.5% | 54.8% | 48.6% |
| **LIQE** | **22/42** | **76.0%** | 59.5% | 90.2% |
| QualiCLIP | 19/42 | 71.9% | 59.5% | 80.6% |
| CLIP-IQA+ | 22/42 | 70.7% | 50.0% | 74.3% |

**Table S21** Complete-case method screening. Sensitivity reports the within-reference mean and number of KADID types above .50. Generalization is the lowest signed $\rho^*$ over all unseen databases. Floor is normalized agreement with Q-Align on the common 42 clusters and its topic-block lower bound (L). Sup.: K = KonIQ-10k opinion training; L = LIVE; OU = opinion-unaware; – = no opinion training; M = multi-database mixture. Q-Align is a reference, not a candidate.

| Method | Sup. | Pol. | Sensitivity | Lowest unseen $\rho^*$ | Floor; L | Eligible |
|---|---|---|---|---|---|---|
| **LIQE** | K | ↑ | .807; 22 | .723 | .551; .284 | **yes** |
| **QualiCLIP** | OU | ↑ | .849; 22 | .665 | .371; .143 | **yes** |
| **CLIP-IQA+** | K | ↑ | .808; 22 | .654 | .236; .034 | **yes** |
| TOPIQ | K | ↑ | .721; 20 | .511 | .506; .289 | no |
| MUSIQ | K | ↑ | .733; 20 | .556 | .551; .297 | no |
| DBCNN | K | ↑ | .668; 19 | .484 | .461; .274 | no |
| HyperIQA | K | ↑ | .673; 19 | .462 | .371; .122 | no |
| NIQE | OU | ↓ | .607; 19 | .368 | .281; .012 | no |
| BRISQUE | L | ↓ | .657; 19 | .236 | .371; .136 | no |
| CLIP-IQA | OU | ↑ | .645; 20 | .501 | .371; .045 | no |
| SIFT+BoVW (deployed) | – | ↑ | −.077; 5 | −.445 | .056; −.308 | no |
| Q-Align | M | ↑ | .880; 23 | – | – | reference |

**Table S22** Two published expressions against the property the same paragraph claims for them. **Above**: the contrast term of the Qaroush visual-attention objective, over a sweep in which contrast rises by construction (25 levels, seed 42). Histogram equalisation drives every frame towards the same output standard deviation, so the published difference measures distance from maximum contrast: it runs backwards, and so does the signed form. Only the ratio, which the equation does not write, agrees. **Below**: the brightness term of the Korolkov module, over a sweep in which exposure rises by construction (21 levels). The source states that it "penalizes under- and over-exposed frames", which requires a non-monotone function; the mean luminance it specifies is monotone and enters the module with weight +0.3, so a brighter frame always scores higher. **No corpus is involved: both inversions are properties of the expressions.**

| Reading | Expression | $\rho$ with the property |
|---|---|---|
| *Qaroush contrast, against contrast (Eq. 8)* | | |
| Eq. 8 as published | $\frac{1}{K}\sum_k \lvert\sigma_p(\mathrm{HE}(f_k)) - \sigma_p(f_k)\rvert$ | −1.0000 |
| without the absolute value | $\frac{1}{K}\sum_k \big(\sigma_p(\mathrm{HE}(f_k)) - \sigma_p(f_k)\big)$ | −1.0000 |
| ratio, unwritten in the source | $\frac{1}{K}\sum_k \sigma_p(f_k)/\sigma_p(\mathrm{HE}(f_k))$ | +1.0000 |
| *Korolkov brightness, against exposure* | | |
| as published | $\frac{1}{\lvert I\rvert}\sum_{x,y} L(I)_{x,y}$ (LAB) | +1.0000 |
| what "penalizes" would need | $-\lvert L - 127.5\rvert$ | −.4883 |

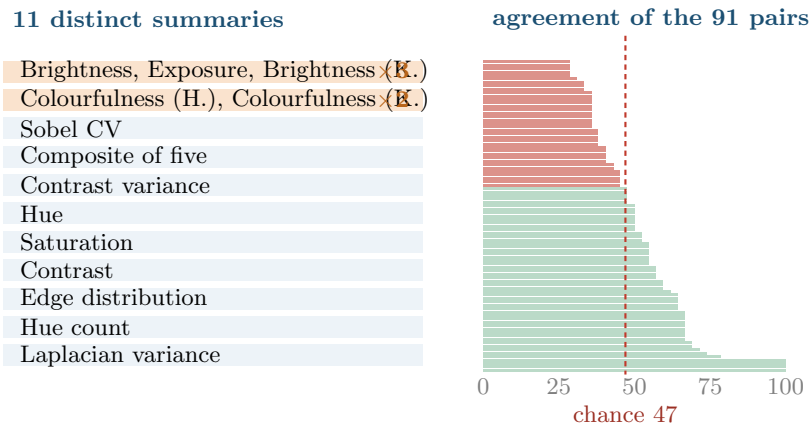



**Fig. S3** The same 42 redundancy clusters, the same pipeline, only the visual-selection operation swapped; no reference model is involved. **Left**: fourteen operations produce eleven distinct summaries, and only two groups agree completely: the three brightness variants and the two colourfulness implementations. **Right**: agreement for each of the 91 operation pairs, sorted, against the chance floor of 47 per cent. Every cluster has at least two operations choosing differently, the average cluster draws 2.21 distinct choices, and **37 of the 91 pairs agree below chance**. The choice of operation is not a detail of the stage; it is most of what the stage decides.

**Table S23** Traceability of every audited operation: the source work, the formula as the source states it, the function that computes it, and the reference model used on the in-domain column. Every operation name and count printed elsewhere in this article is derived from this table. **Dir.** is the direction the source declares; spatial information declares none, which is why it is measured but carries no verdict.

| Operation | Source | Formula, as the source states it | Function | Dir. | Ref. |
|---|---|---|---|---|---|
| Averaged aesthetics (3) | [2] | unweighted mean of colourfulness, contrast, edge distribution | `gygli_media_crua` | ↑ | LIQE |
| Colourfulness (Hasler) | [3] | $M^{(3)} = \sigma_{rgyb} + .3\mu_{rgyb}$ | `hasler_colorfulness` | ↑ | LIQE |
| Colourfulness (Ke) | [7] | Hasler $M^{(3)}$; Ke gives no formula | `hasler_colorfulness` | ↑ | LIQE |
| Composite of five | [10], [7] | unweighted mean of the five Ke terms | `srinivas_quality` | ↑ | LIQE |
| Contrast | [7] | width of central 98% of the grey histogram (Ke §4.5) | `ke_contraste` | ↑ | LIQE |
| Contrast variance | [6], [11] | pixel variance | `contraste_variancia` | ↑ | LIQE |
| Edge distribution | [7] | $q_a = 1 - w_x w_y$ (Ke §4.1) | `ke_edge_distribution` | ↑ | LIQE |
| Entropy | [9] | Shannon entropy over HSV histogram bins (Eq. 5) | `qaroush_entropia` | ↑ | LIQE |
| GLCM contrast | [12] | GLCM contrast descriptor | `glcm_contraste` | ↑ | LIQE |
| Gradient magnitude | [9] | mean Sobel gradient magnitude (Eq. 10) | `qaroush_gm` | ↑ | LIQE |
| Gradient orientation | [9] | SD of Sobel gradient orientations (Eq. 15) | `qaroush_go` | ↑ | LIQE |
| Hue count | [7] | $q_h = 20 - \lvert N \rvert$, $\alpha = .05$ (Ke §4.3) | `ke_hue_count` | ↑ | LIQE |
| Laplacian variance | [1], [8] | variance of the grey-scale Laplacian | `laplacian_var` | ↑ | LIQE |
| Saliency area | [9] | area fraction above a saliency threshold (Eq. 6) | `qaroush_saliencia` | ↑ | LIQE |
| Saturation | [4] | mean HSV saturation (§III-A-1) | `huang_saturacao` | ↑ | LIQE |
| Spatial information | [5] | Sobel energy, ITU-T P.910 | `si` | none | LIQE |

All sixteen are computed by `ops_da_literatura.py` on the same frames and pooled identically. **Ref.** is the model each operation is compared against on the in-domain column; the ten no-reference models in the main screening table use Q-Align there instead, so that column compares each family with its own reference. Colourfulness has no formula in Ke, who cites Datta; the Hasler expression is used and the substitution is declared here rather than silently. Scores for fifteen operations are in `corpus_inteiro.json`; spatial information is scored on the extended 83-segment design by `si_no_corpus_de_video`, in `designA_estendido_83.json`.

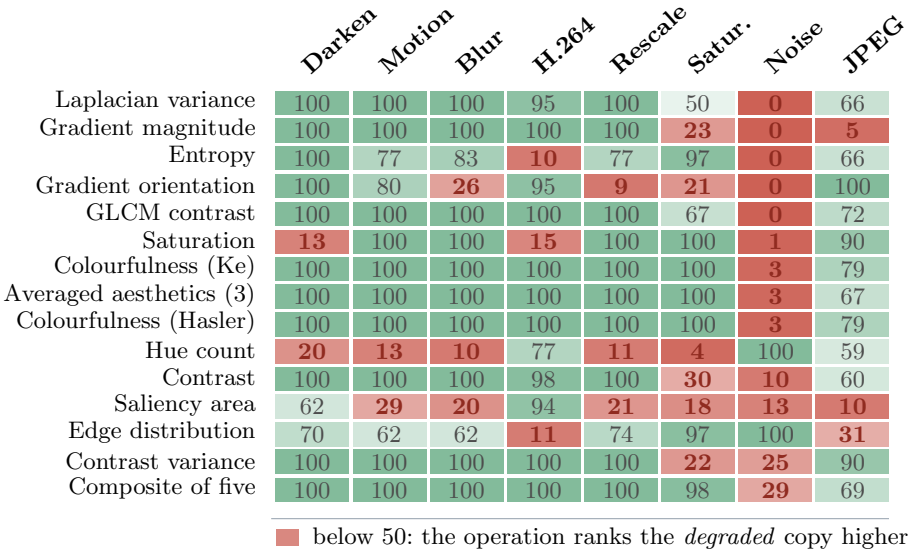



**Fig. S4** Paired accuracy of each summarization heuristic on 800 unmodified–degraded broadcast pairs. Each cell is 100 pairs, 20 segments at five severity levels, and the better member is fixed by construction. **Every operation falls below chance on at least one type, and thirteen of the fifteen do so under white noise**, five of them on every pair. Rows are ordered by worst cell.

**Table S24** The three families under the screening rule of Section 3.5 of the main text, whose three conditions are stated there. **Each cell is the maximum over its family on that axis alone**, and the two maxima need not belong to the same member: among the no-reference models the best sensitivity is QualiCLIP and the best generalization LIQE. Spatial information declares no direction and the reproduced modules were measured on the constructed pairs rather than on these axes, so neither carries a verdict here; the heuristics row counts 16 computed and 15 evaluated. Per-operation values are in Table S18 and per-model values in Table S21.

| Family | $n$ | Best sens. | Best gen. | Clears both |
|---|---|---|---|---|
| Summarization heuristics | 16 | +.298 | +.139 | 0 of 15 evaluated |
| Published composites | 19 | +.524 | +.067 | 0 of 19 (1 clears sens.) |
| Reproduced modules | 2 | — | — | not screened |
| No-reference models | 10 | +.849 | +.723 | **3 of 10** |
| Deployed SIFT+BoVW | 1 | −.077 | −.445 | 0 of 1 |

**Table S25** Origin, documented use and evaluated object for the cases that carry the claims here. **A citation does not say which work used the operation in selection, nor what this bench measured**, so the three are separated; where origin and use fall in the same work the middle column says so, and the last column marks our own adaptations as such. The full inventory, with formulas and results files, is in Tables S17 and S23.

| Operation | Origin and role there | Documented use in selection | Evaluated here |
|---|---|---|---|
| Edge distribution, hue count, contrast | Ke et al. [7]: features of a naive Bayes classifier for photo quality | Srinivas et al. [10] cites Ke for all three (§3.1.3–3.1.5) and averages them into a per-frame quality score | each term alone, plus the mean and an adapted classifier |
| Colourfulness in that same score | Swain and Ballard, by histogram intersection, as Srinivas et al. [10] states | Srinivas et al. [10], §3.1.1 | the Hasler expression instead, a declared substitution |
| Sharpness and brightness | Korolkov [8]: two terms of a keyframe score | the same work, $0.7\,z(\text{sharp}) + 0.3\,z(\text{bright})$ | the published rule, with our normalization population stated |
| Contrast variance | Jeong et al. [6]: a badness term inside a cost multiplicative in cluster distance | the same work, within that objective | the isolated statistic, read in the increasing direction |
| Gradient magnitude, saliency, entropy, gradient orientation | Qaroush et al. [9]: terms of four numbered objectives | the same work; two of the four objectives are not about technical quality | each term alone, and the visual-attention module of Eq. 20 |

**Table S26 What each source work reports as evidence for its own visual criterion.** Read from the full text of each paper; the excerpt is verbatim. *Tests* names what the source puts to a test, which is not always the operation the later literature borrows. Two sources are not in our archive and therefore carry no verdict: **absence of a reading is not absence of validation**. It documents the clause carried on each band of the audit table in the main text.

| Source | Tests | What it reports | Verbatim excerpt |
|---|---|---|---|
| Ke et al. [7] | **human judgment of the property** | community ratings on 60,000 photos as ground truth; per-feature precision–recall and a combined classifier at 27.8 per cent equal error, on the extremes only | *"the photos have been rated by its community of users, giving us ground truth on the quality of each photo . . . The top and bottom 10% of the photos were extracted . . . we chose to ignore the middle 80%"* |
| Hasler and Süsstrunk [3] | **human judgment of the property** | a psychophysical scaling experiment, seven categories, the expression fitted to the ratings at 94.0 per cent; the property is colourfulness | *"We set up a psychophysical category scaling experiment, and ask people to rate images using 7 categories of colourfulness. We then fit a metric to the results"* |
| Srinivas et al. [10] | the summary only | precision, recall and F of the selected keyframes against the ground-truth keyframes of seven open-video.org videos | *"results are compared with the ground truth keyframes provided by open-video.org"* |
| Qaroush et al. [9] | the summary only | precision, recall and F1 of the extracted segments against reference summaries | *"The evaluation focuses on comparing segments extracted by our system with those identified in ground truth summaries"* |
| Korolkov [8] | **neither the rule nor its weights** | weights set, not fitted; ablations vary segmentation parameters, and keyframes are judged by coverage and a qualitative review | *"We set wsharp = 0.7, wbright = 0.3 to favor focused frames"* |
| Cakmak et al. [1] | the summary, with an ablation | TVSum20 and SumMe against human importance scores, with an ablation removing each modality | *"Removing any single component"* |
| Yang et al. [12] | the summary, with an ablation | structural and feature ablations scored by F on SumMe over 15 annotations | *"we conducted experiments on the SumMe dataset for structural ablation of the model and for feature ablation"* |
| Gygli et al. [2] | the summary, with an ablation | f-measure against multiple human-created summaries per video, collected in a study with perception psychologists, plus an ablation measuring the gain of adding each feature to the set | *"We evaluate our method using multiple 'ground truth' summaries per video, which we acquired in a study in collaboration with perception psychologists"* |
| Jeong et al. [6] | the summary only | mean opinion scores from human subjects and precision/recall against labeled events; no ablation or isolated test of the contrast term | *"the visual quality is the combination of contrast, blur, and image skew measures"* |
| Huang et al. [4] | the summary only | Spearman and Kendall rank correlation against human annotations on TVSum and SumMe | *"rank correlation coefficients measure how well the ordering"* |

**Table S27** Resampling unit per analysis, and the rival unit tested against it. Observations are nested, so the unit decides the width of every interval: the italic line under each row states the nesting. Each interval was recomputed under the declared unit and under the rival, and the declared unit is accepted only when the rival fails to reproduce the published bounds. One of the four did not pass and was corrected.

| Block | Analysis | Declared unit | Rival tested | Outcome |
|---|---|---|---|---|
| Block 01 | Paired difference, 800 pairs | segment block | the pair | **the published interval matched the rival scheme; corrected to** $[+29.8, +38.9]$ |
| | *20 segments, 40 pairs each; pairs within a segment share content* | | | |
| Block 01 | Strict accuracy, 800 pairs | topic block | the segment block | reproduces under the declared unit and not under the rival |
| | *the 20 segments come from 13 news topics* | | | |
| Block 04 | In-domain floor, 42 clusters | topic block | the cluster | reproduces under the declared unit and not under the rival |
| | *the 42 clusters come from 19 topics* | | | |
| Block 04 | Output contrast, 42 clusters | topic block | the cluster | reproduces under the declared unit and not under the rival |
| | *the 42 clusters come from 19 topics* | | | |

**Table S28** Every operation–type cell below the 50 per cent chance floor, with the accuracy of each. All 35 cells are significant by an exact one-sided binomial test under Holm correction across the 120 operation–type tests; a cell short of significance would carry † and none does. Counting ties inside the denominator instead of outside leaves the same 35 cells and the same 15 operations. Removing the two types aligned with an operation's own attribute, darkening and desaturation, leaves all 15 operations still inverting; removing white noise instead leaves 9. Dropping one segment at a time, all 15 keep inverting in every one of the 20 rounds. Dk = darkening, Mo = motion blur, Bl = Gaussian blur, H2 = H.264, Rs = rescaling, St = desaturation, Ns = white noise, Jp = JPEG.

| Operation | Types | Worst | Cells below chance (%) |
|---|---|---|---|
| Saliency area | 6 | 10.0 | Mo 29.0, Bl 20.0, Rs 21.0, St 18.0, Ns 13.1, Jp 10.0 |
| Hue count (Ke) | 5 | 4.2 | Dk 21.7, Mo 15.7, Bl 12.3, Rs 14.1, St 4.2 |
| Gradient orientation | 4 | 0.0 | Bl 26.0, Rs 9.0, St 21.0, Ns 0.0 |
| Gradient magnitude | 3 | 0.0 | St 23.0, Ns 0.0, Jp 5.0 |
| Saturation | 3 | 1.0 | Dk 13.0, H2 15.0, Ns 1.0 |
| Entropy | 2 | 0.0 | H2 10.0, Ns 0.0 |
| Contrast (Ke) | 2 | 10.3 | St 31.6, Ns 10.3 |
| Edge distribution | 2 | 11.0 | H2 11.0, Jp 31.0 |
| Contrast variance | 2 | 22.0 | St 22.0, Ns 25.0 |
| GLCM contrast | 1 | 0.0 | Ns 0.0 |
| Laplacian variance | 1 | 0.0 | Ns 0.0 |
| Gygli mean of three | 1 | 3.0 | Ns 3.0 |
| Colourfulness (Hasler) | 1 | 3.0 | Ns 3.0 |
| Colourfulness (Ke) | 1 | 3.0 | Ns 3.0 |
| Srinivas composite | 1 | 29.0 | Ns 29.0 |

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